CLICK HERE CLICK HERE

APUL

NEET FLIX

AIM FREE EDUCATION TO ALL

NEET CHANNEL



JOIN OUR TELEGRAM CHANNELS HERE

TEST PAPERS <<< CLICK

TOPPER'S NOTES <<< CLICK

BEST FROM KOTA LECTURES <<< CLICK

QUIZES <<< CLICK

REVISION FLASH CARDS &

MANY MORE <<< CLICK

MM : 720

TEST - 5

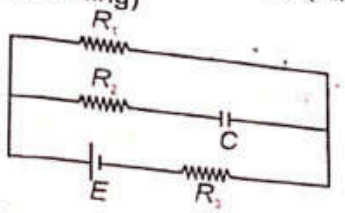
Time : 3 Hrs. 20 Min.

[PHYSICS]

Choose the correct answer :

SECTION - A

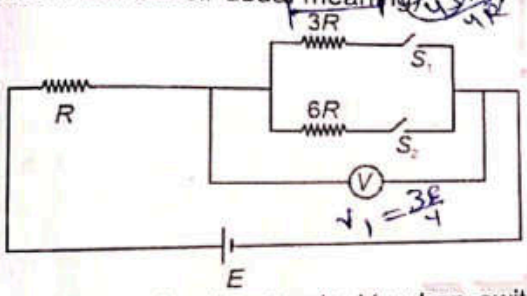
1. The charge on the capacitor in steady state for the electrical network shown below is (All symbols have their usual meaning)



(1) $\frac{CER_3}{R_1+R_2}$ (2) $\frac{CER_2}{R_1+R_3}$

(3) $\frac{CER_1}{R_1+R_3}$ (4) $\frac{CER_2}{R_1+R_2}$

2. An electrical circuit is shown in figure. (All the symbols have their usual meaning)



The reading of voltmeter is V_1 when switch S_1 is closed and S_2 is kept open, while it is V_2 when switch S_2 is closed and S_1 is kept open.

Based upon the above information which option is correct?

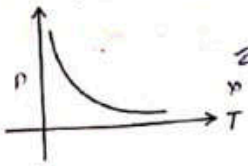
(1) $V_1 > V_2$

(2) $V_1 < V_2$

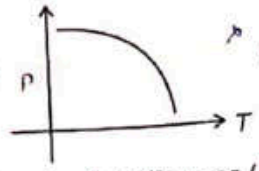
(3) $V_1 = V_2$

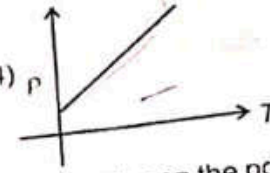
(4) Insufficient information

3. Which of the following graph represents the variation of resistivity (ρ) with temperature (T) for nichrome?

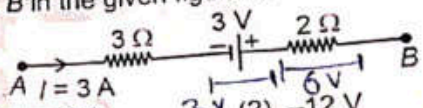
(1) 

(2) 

(3) 

(4) 

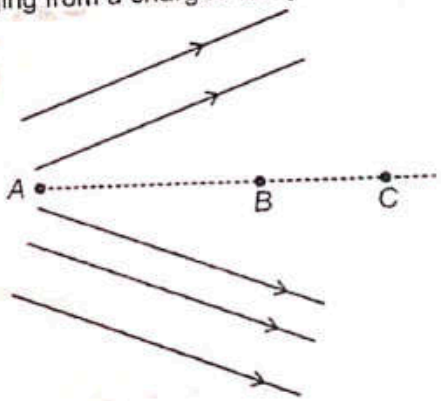
4. The potential difference ($V_A - V_B$) between the points A and B in the given figure is



(1) 12 V (2) −12 V

(3) 9 V (4) −9 V

5. Figure shows some of the electric lines of force emerging from a charged body.



Space for Rough Work

If F_A, F_B and F_C are the forces experienced by a test charge at positions A, B and C respectively, then

(1) $F_A > F_B > F_C$ (2) $F_A = F_B = F_C$

(3) $F_A < F_B < F_C$ (4) $F_A > F_B = F_C$

Consider the following statements out of which one is labelled as assertion and the other as reason.

Assertion (A): When two like charges are brought near to each other, their electrostatic potential energy increases.

Reason (R): The concept of potential holds true only for conservative fields.

In the light of above statements, choose the correct answer.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) Both (A) and (R) are false

7. Charge crossing through a cross-section of a conductor is given by $q = (t^3 + 2t)$ coulomb. The instantaneous current through the conductor $t = 1$ s is

(1) 2 A (2) 5 A

(3) 7 A (4) 10 A

8. The S.I. unit of mobility is (All symbols have their usual meaning)

(1) $\frac{m^2 s}{V}$ (2) $\frac{m^2 V}{s}$

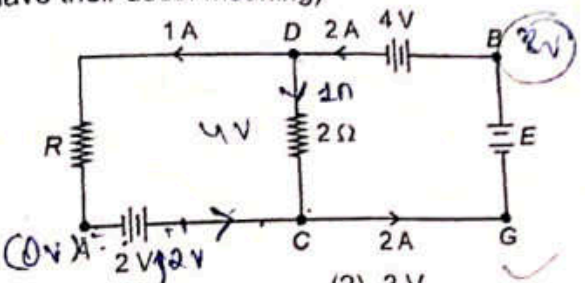
(3) $\frac{V s}{m^2}$ (4) $\frac{m^2}{V s}$

When a potential difference V_0 is applied across a wire of length L, the drift speed of electrons is v_d. If the electron density in the wire is n, then the resistivity of material of the wire is equal to

(1) $\frac{V_0}{neLv_d}$ (2) $\frac{v_d}{nel\,V_0}$

(3) $\frac{V_0^2}{neLv_d}$ (4) $\frac{v_d^2}{ne^2LV_0}$

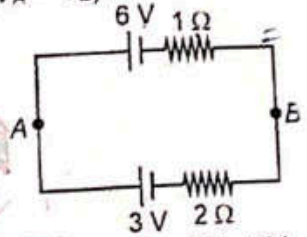
10. In the given circuit, assuming point A to be at zero potential, the potential at point B is (All the symbols have their usual meaning)



(1) 2 V (2) 3 V

(3) 1 V (4) 8 V

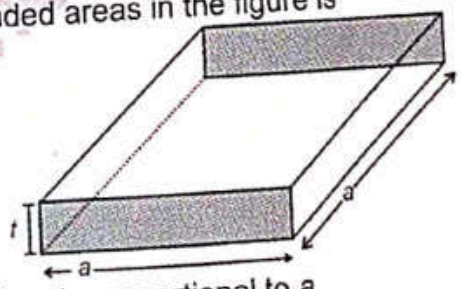
11. Two batteries of different emfs and different internal resistances are connected as shown. The potential difference ($V_A - V_B$) is



(1) 4.5 V (2) 5 V

(3) 7 V (4) 6.5 V

12. Consider a thin sheet having two sides of length a and thickness t, made of a material of resistivity ρ. The resistance between two opposite faces, shown by shaded areas in the figure is



(1) Directly proportional to a

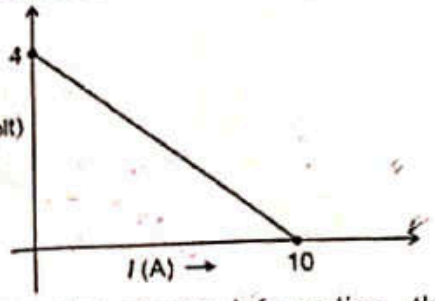
(2) Directly proportional to t

(3) Independent of a

(4) Independent of t

Space for Rough Work

13. For a cell, the graph between the potential difference (V) across the terminals of the cell and the current (I) drawn from the cell is shown in figure.



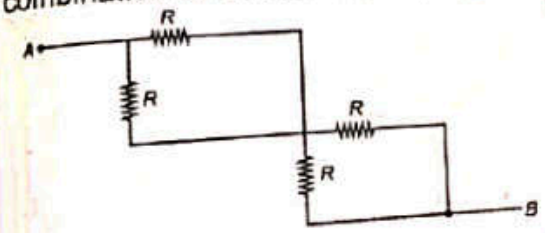
Based upon the above information, the internal resistance of the cell is

(1) 0.4 Ω (2) 2 Ω
(3) 0.25 Ω (4) 1.3 Ω

14. An electric bulb marked 50 W, 100 V is connected across a 25 V supply. The power consumed by the electric bulb will be

(1) 25 W (2) 2.5 W
(3) 3.125 W (4) 6.25 W

15. The equivalent resistance across points A and B for the combination of resistors as shown below, is



(1) R (2) $\frac{R}{2}$
(3) $2R$ (4) $\frac{R}{4}$

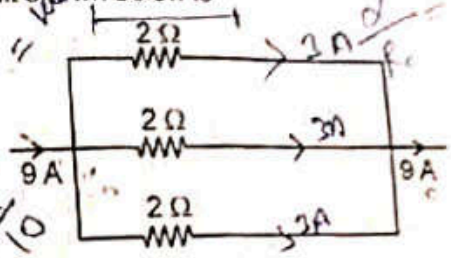
16. A wire of resistance R_0 is drawn out, so that its length is increased by hundred percent of its original length. If its new resistance becomes 20 Ω, then the value of R_0 will be .

(1) 10 Ω (2) 15 Ω
(3) 20 Ω (4) 5 Ω

17. The equivalent resistance of n equal resistors connected in parallel is α. When those n resistors are connected in series, then the equivalent resistance will become

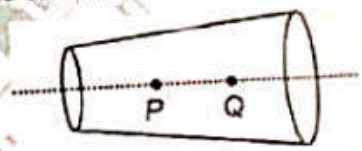
(1) $\frac{\alpha}{n}$ (2) $\frac{\alpha}{n^2}$
(3) $n^2\alpha$ (4) $n\alpha$

18. The power dissipated in each resistance in the circuit shown below is



(1) 12 W (2) 18 W
(3) 24 W (4)

19. A current-carrying wi[illegible]-uniform cross-section is shown in



A steady current flows through it. The magnitude of current density at P and Q is λ_P and λ_Q. Based on the above information, which of the given option is correct?

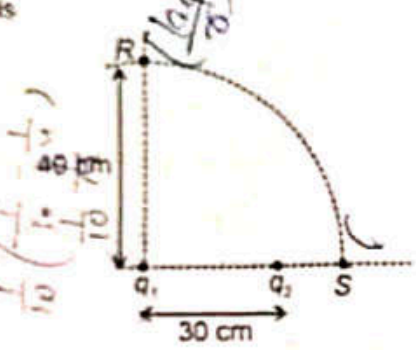
(1) $\lambda_P = \lambda_Q$ (2) $\lambda_P > \lambda_Q$
(3) $\lambda_P < \lambda_Q$ (4) Insufficient information

20. A charge Q lies at the centre of a Gaussian spherical surface of radius R. If the radius is halved, then the net electric flux will

(1) Be halved (2) Become twice
(3) Remain same (4) Become four times

Space for Rough Work

21. Two charges q_1 and q_2 are placed 30 cm apart as shown. A unit positive point charge is moved along the arc of circle of radius 40 cm from R to S. The work done in moving this unit positive charge from R to S is



(1) Zero (2) $\frac{2q_2}{\pi\varepsilon_0}$

(3) $\frac{q_2}{4\pi\varepsilon_0}$ (4) $\frac{2q_1}{4\pi\varepsilon_0}$

22. A point charge Q is revolving around a fixed charge q in a circle of radius R. The work done by electrostatic force on charge Q in one complete revolution will be

(1) Zero (2) $Q \times 2\pi \times R$

(3) $\frac{q \times R \times kQ}{2R^2}$ (4) $\frac{2\pi kqR}{R^2}$

23. The potential due to an electric dipole of dipole moment p at a distance r from centre of dipole on its axis, is

(1) $\frac{p}{4\pi\varepsilon_0 r^2}$ (2) Zero

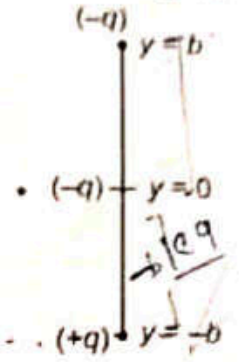
(3) $\frac{-p}{4\pi\varepsilon_0 r^2}$ (4) $\frac{p}{2\pi\varepsilon_0 r^2}$

24. The correct dimensional formula for electrostatic potential is

(1) $[ML^2T^{-3}A^{-1}]$ (2) $[ML^{-2}T^3A^{-1}]$

(3) $[ML^{-2}T^{-3}A^1]$ (4) $[ML^2T^{-3}A^2]$

25. Three charges are placed along y-axis at $y = -b$, $y = 0$ and $y = b$ as shown in figure.

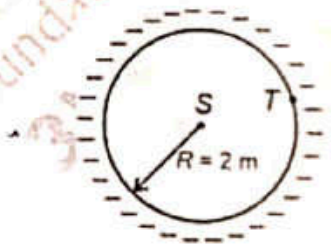


The electrostatic potential energy of the system is

(1) $\frac{-1}{4\pi\varepsilon_0}\left(\frac{q^2}{2b}\right)$ (2) $-\frac{1}{4\pi\varepsilon_0}\left(\frac{q^2}{b}\right)$

(3) $-\frac{1}{4\pi\varepsilon_0}\left(\frac{q^2}{4b}\right)$ (4) Zero

26. A thin spherical shell is uniformly charged by charge $q = -3\mu C$. The magnitude of potential difference between two points S and T (in V) shown in the figure is



$q = -3\ \mu C$

(1) 3×10^5 (2) 1×10^5

(3) 1.5×10^5 (4) Zero

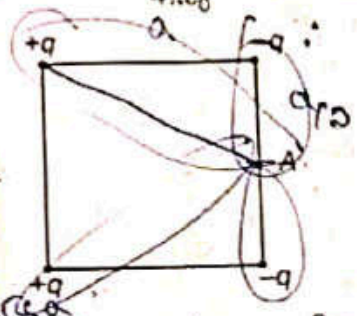
27. An electric dipole when placed in a uniform electric field has minimum electrostatic potential energy. The angle between dipole moment and electric field is

(1) 90° (2) 180°

(3) 0° (4) 270°

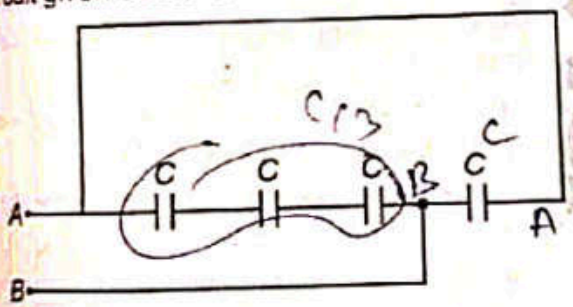
29. In a certain region of space with volume 0.3 m^3, the electric potential is found to be 6 V throughout. The magnitude of electric field in this region is
(1) 0.5 N/C (2) 5 N/C
(3) 3 N/C (4) Zero

30. Four electric charges $+q$, $+q$, $-q$ and $-q$ are placed at corners of a square of side 2 m. The electric potential at point A midway between the two $-q$ charges is (where $k = \frac{1}{4\pi\varepsilon_0}$)



(1) Zero (2) $2kq\left[-1+\frac{1}{\sqrt{5}}\right]$
(3) $2kq\left[1-\frac{1}{\sqrt{5}}\right]$ (4) $2kq\left[1+\frac{1}{\sqrt{5}}\right]$

31. The equivalent capacitance between A and B in circuit given below is



(1) $\frac{C}{4}$ (2) $\frac{3C}{4}$
(3) $\frac{4C}{3}$ (4) $\frac{C}{2}$

32. The charge on α-particle is equal to
(1) 3.2×10^{-19} C (2) 1.6×10^{-19} C
(3) 4.8×10^{-19} C (4) 2.3×10^{-19} C

33. The variation of electric potential (V) with distance x(in metre) is given by $V = (x^2 + 10x - 9)$ volt. The magnitude of electric field at $x = 1$ m is
(1) Zero (2) 12 V/m
(3) 6 V/m (4) 11 V/m

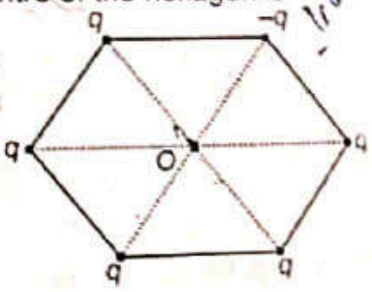
34. Two charges $-2Q$ and $+Q$ are placed as shown in figure.



The point on the line joining the two charges at which electric field intensity is zero, will be
(1) Between +Q and −2Q
(2) On the left of +Q
(3) On right of −2Q
(4) On the perpendicular bisector of line joining charges

35. Electric field in a region is uniform and is given by $\vec{E} = a\hat{i} + b\hat{j}$. The electric flux associated with a surface of area $\vec{A} = \pi R^2 \hat{j}$ is
(1) πaR^2 (2) πbR^2
(3) Zero (4) $\pi(a + b)R^2$

36. Six point charges are placed at the vertices of a hexagon of side 1 m as shown in figure. Net electric field at centre of the hexagon is



(1) Zero (2) $\frac{6q}{4\pi\varepsilon_0}$
(3) $\frac{2q}{4\pi\varepsilon_0}$ (4) $\frac{q}{4\pi\varepsilon_0}$

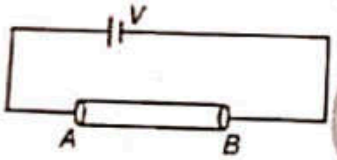
Space for Rough Work

SECTION-B

36. Dielectric constant for a metal is
(1) Zero (2) Infinite
(3) 1 (4) 10

37. A particle having charge q_1 exerts an electrostatic force F_0 on charge q_2 at rest. If a particle having charge 5e is placed midway on the line joining the two charges q_1 and q_2, then electrostatic force on q_2 due to q_1 will
(1) $2F_0$ (2) $\frac{F_0}{2}$
(3) $\frac{F_0}{5}$ (4) F_0

38. When a battery of emf V is applied across a wire AB, the drift speed of electrons through the conductor is v_1. If the battery is replaced by another battery of emf $2V$, then new drift speed of electrons will be

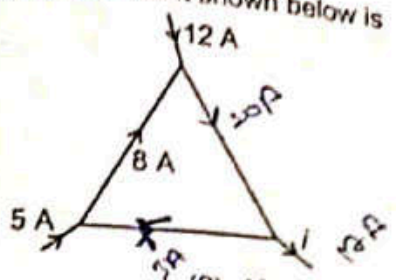


(1) $2v_1$ (2) v_1
(3) $\frac{v_1}{2}$ (4) $3v_1$

39. If current through a conductor at any instant is $i = (2 + 3t)$ A, then the charge transferred between $t = 0$ to $t = 2$ s is
(1) Zero (2) 5 C
(3) 15 C (4) 10 C

40. A current of 4 A is to be sent through a load resistor of 5 Ω. The number of cells required to be connected in series, if each has emf 3 V and internal resistance 0.5 Ω, is
(1) 40 (2) 30
(3) 20 (4) 10

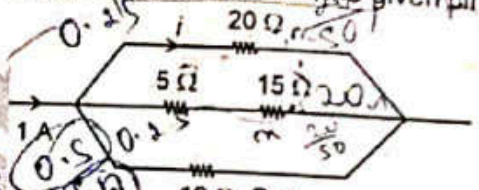
41. The current I in the circuit shown below is



(1) 17 A (2) 13 A
(3) 7 A (4) 10 A

42. If the length of a conductor is tripled, then resistivity will
(1) Be doubled
(2) Be halved
(3) Be tripled
(4) Remain same

43. The magnitude of current i in the given circuit is



(1) 0.5 A (2) 1 A
(3) 0.25 A (4) 2 A

44. Read the following statements and choose the correct option.
(a) The resistivity of a semiconductor decreases with increase in temperature.
(b) The average time of collisions decreases with increase in temperature.
(c) Resistance of a conductor is independent of its dimensions.
(1) Only (a) and (b) are correct
(2) Only (b) and (c) are correct
(3) Only (c) is correct
(4) All (a), (b) and (c) are correct

Space for Rough Work

Consider the following statements:

Statement-I: When an uncharged capacitor is charged by a battery, only 50% of the energy supplied by the battery is stored in the capacitor.

Statement-II: The total potential difference applied across two capacitors in series will be distributed in inverse ratio of capacitance of the capacitors.

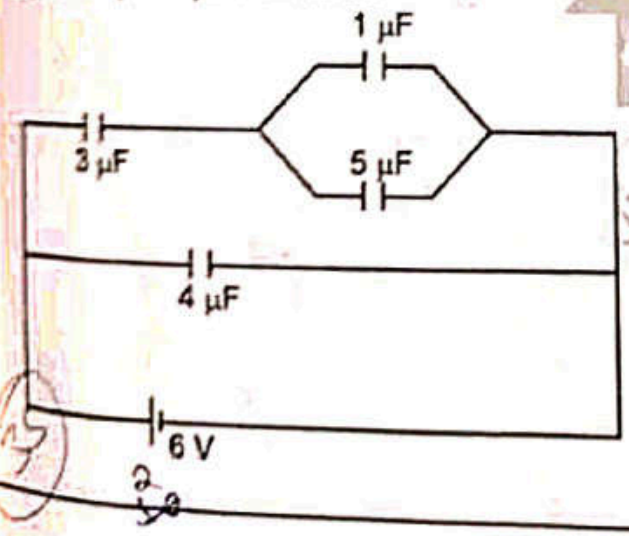
In the light of the above statements, choose the correct answer.

(1) Both statement I and statement II are correct
(2) Both statement I and statement II are incorrect
(3) Statement I is correct while statement II is incorrect
(4) Statement I is incorrect while statement II is correct

46. A capacitor of capacitance C is charged to a potential difference V from a cell and then disconnected from it. A charge $+Q$ is now given to its positive plate. The potential difference across the capacitor now is

(1) V
(2) $V+\frac{Q}{2C}$
(3) $V-\frac{Q}{2C}$
(4) Zero

47. In the circuit shown in figure, the ratio of charges on 3 μF to 4 μF capacitor will be

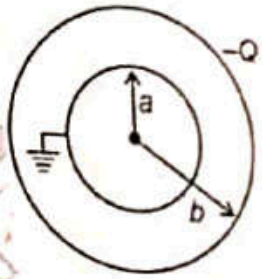


(1) $\frac{1}{8}$
(2) $\frac{1}{4}$
(3) 4
(4) $\frac{1}{2}$

48. A pendulum bob of mass m carrying a charge q is at rest with its string making an angle β with the vertical in a uniform horizontal electric field E. The tension in the string is

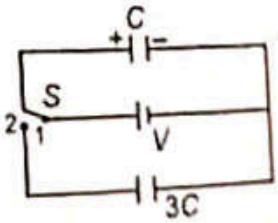
(1) $\frac{Mg}{\sin\beta}$
(2) $\frac{qE}{\cos\beta}$
(3) $\frac{qE}{\sin\beta}$
(4) qE

49. Two concentric hollow metal shells have radii a and b. The outer shell is given a negative charge $-Q$ and the inner shell is earthed. The charge on inner sphere will be



(1) $\frac{Qb}{a}$
(2) $\frac{Qa}{b}$
(3) Zero
(4) $\frac{Qa}{2b}$

50. A capacitor C is charged by a battery having emf V. Then capacitor C is disconnected from the battery and connected to an uncharged capacitor of capacity $3C$ by shifting switch S from 1 to 2 as shown in figure.



Space for Rough Work

In Column-I certain quantities are given and their values are given in Column-II. Match the correct entries in Column-I with corresponding correct entries in Column-II

	Column-I		Column-II
a.	After shifting switch S from 1 to 2, the final energy stored in C	(p)	$\frac{3CV^2}{8}$
b.	After shifting switch S from 1 to 2, the energy stored in $3C$	(q)	$\frac{CV^2}{32}$
c.	After shifting S from 1 to 2, the loss of energy during redistribution of charge	(r)	$\frac{3CV^2}{32}$

(1) a(q), b(r), c(p)

(2) a(r), b(q), c(p)

(3) a(q), b(p), c(r)

(4) a(p), b(r), c(q)

[CHEMISTRY]

SECTION-A

The correct increasing order of boiling points of the following alkanes is

a. n-Hexane b. n-Pentane

c. 2-methylbutane d. 2,3-dimethylpropane

Choose the **correct** answer from the following options

(1) a < b < c < d (2) d < b < c < a

(3) d < c < b < a (4) b < d < c < a

Which of the following alkanes cannot be prepared by Kolbe's electrolysis method?

(1) Ethane (2) Butane

(3) Hexane (4) Methane

Consider the following statements:

Statement I : Reaction of $CH_3 - CH = CH - CH_2Cl$ with aqueous KOH gives But-2-en-1-ol as one of the products.

Statement II : Reaction of methyl chloride with hydroxide ion gives methyl alcohol as product, following S_N1 mechanism.

In the light of above statements choose the **correct** option.

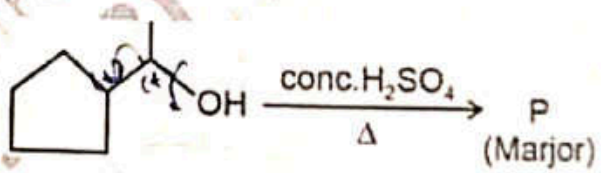
(1) Statement I is correct but statement II is incorrect

(2) Statement I is incorrect but statement II is correct

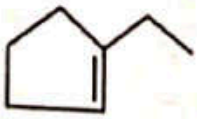
(3) Both statement I and statement II are correct

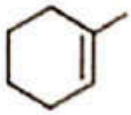
(4) Both statement I and statement II are incorrect

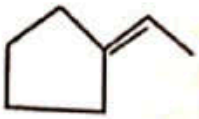
Consider the following reaction



The major product (P) is

(1) 

(2) 

(3) 

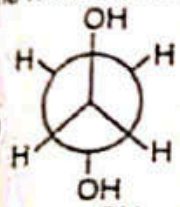
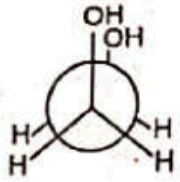
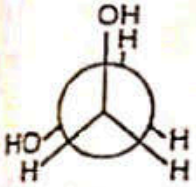
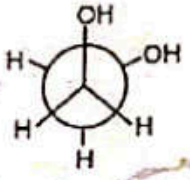
(4) 

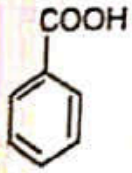
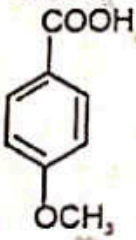
Space for Rough Work

1° SN2.

$CH_3Cl \xrightarrow{OH^-} CH_3OH$

55. Choose the **incorrect** statement.
(1) Cis-but-2-ene is more polar than trans-but-2-ene
(2) Dipole moment of trans-but-2-ene is almost zero
(3) Boiling point of trans-but-2-ene is higher than that of cis-but-2-ene
(4) It is observed that the trans but-2-ene has higher melting point than cis-but-2-ene

56. The most stable conformer of ethane-1-2-diol is
(1) 
(2) 
(3) 
(4) 

57. Which of the given acids has the highest pK_a value?
(1) CCl_3COOH
(2) CF_3COOH
(3) 
(4) 

58. Match the **column I** with **column II** and select the correct option.

	Column I (Reagent)		Column II Reagent name
a.	$CrO_3 - H_2SO_4$	(i)	Fehling reagent
b.	Cold dilute alkaline $KMnO_4$	(ii)	Tollens' reagent
c.	$[Ag(NH_3)_2]NO_3$	(iii)	Baeyer's reagent
d.	Aq.$CuSO_4$ and Rochelle salt (aq) in equal amount	(iv)	Jones Reagent

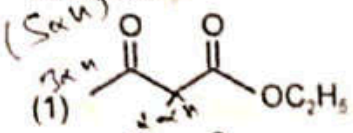
Identify the correct answer from the options given below.
(1) a(iii), b(i), c(iv), d(ii) (2) a(iii), b(iv), c(ii), d(i)
(3) a(iv), b(iii), c(ii), d(i) (4) a(iv), b(iii), c(i), d(ii)

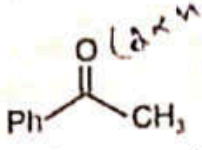
59. Consider the following statements.
Statement I : Carboxylic acids are reduced to primary alcohols by lithium aluminium hydride or with diborane.
Statement II: Sodium borohydride does not reduce the carboxylic acid.
In the light of above statements, choose the **correct** option.
(1) Statement I is incorrect but Statement II is correct
(2) Both statement I and statement II are correct
(3) Both statement I and statement II are incorrect
(4) Statement I is correct but statement II is incorrect

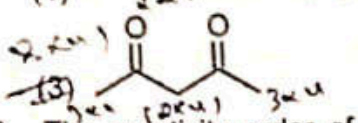
60. Given below are the two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A) : Benzoic acid does not undergo Friedel-Crafts reactions.
Reason (R) : Carboxylic acid group of benzoic acid is deactivating group towards electrophilic substitution and it gets bonded to aluminium chloride during Friedel – Crafts reaction.
In the light of above statement, choose the most appropriate answer from the options given below.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(3) (A) is true but (R) is false
(4) Both (A) and (R) are false

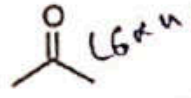
Space for Rough Work

61. The compound with highest percentage of enol content is

(1) 

(2) 

(3) 

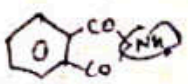
(4) 

62. The reactivity order of halogens with ethane in halogenation reaction is

(1) $Cl_2 > Br_2 > I_2 > F_2$ (2) $F_2 > Cl_2 > Br_2 > I_2$

(3) $F_2 > Br_2 > Cl_2 > I_2$ (4) $I_2 > F_2 > Cl_2 > Br_2$

63. Consider the following reaction sequence

Phthalic acid + $NH_3 \rightleftharpoons$ A $\xrightarrow[-H_2O]{\Delta}$  B $\xrightarrow[\text{Strong Heating}]{-NH_3}$ C

Product C is

(1) Phthalic anhydride (2) Phthalamide

(3) Phthalimide (4) Phthalic acid

64. Consider the following statements

(i) Cannizzaro reaction of formaldehyde is an example of disproportionation reaction.

(ii) Benzaldehyde undergoes Cannizzaro reaction to form salt of benzoic acid and phenol.

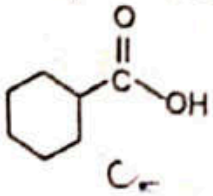
(iii) In Cannizzaro reaction, aldehyde in which α - hydrogen is absent reacts with dilute alkali.

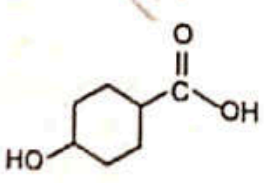
Choose the **incorrect** statement(s)

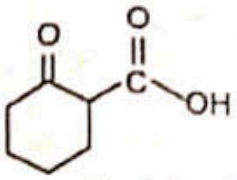
(1) (i) and (ii) only (2) (ii) and (iii) only

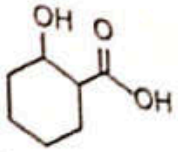
(3) (iii) only (4) (i), (ii) and (iii)

65. Which of the following compounds can be decarboxylated most easily on heating?

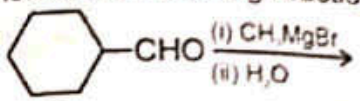
(1) 

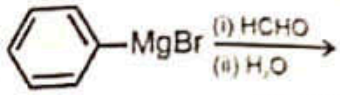
(2) 

(3) 

(4) 

66. Consider the following reactions:

a. 

b. 

c. $CH_3COOC_2H_5 \xrightarrow[(ii)H_2O]{(i)CH_3MgBr}$

In which of the above reactions, at least one of the products is a primary alcohol?

(1) (a) and (b) only (2) (b) and (c) only

(3) (a) only (4) (b) only

67. An alkane 'X' gives three monochloro products on chlorination in diffused sunlight. Alkane 'X' is

(1) n-Butane (2) Propane

(3) Neopentane (4) Isobutane

68. The pair(s) of compounds which cannot be distinguished by iodoform test is

(i) Benzophenone and ethanol

(ii) Acetone and acetophenone

(iii) Ethanal and butanone

Choose the **correct** option.

(1) (ii) only (2) (ii) and (iii) only

(3) (i) and (iii) only (4) (iii) only

69. Which of the following will give fastest S_N1 reaction?

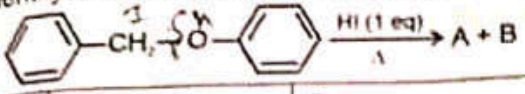
(1) 

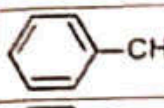
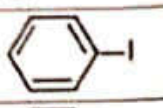
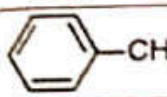
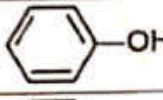
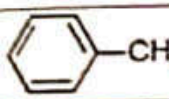
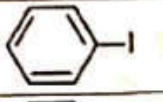
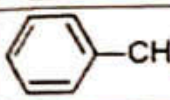
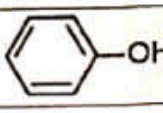
(2) $CH_2 = CH - CH_2Cl$

(3) $CH_3 - \underset{Cl}{\underset{|}{CH}} - CH_3$ (4) $CH_3 - \underset{CH_3}{\underset{|}{\overset{Cl}{\overset{|}{C}}}} - CH_3$

Space for Rough Work

Identify **A** and **B** in the given reaction



	A	B
(1)		
(2)		
(3)		
(4)		

Consider the following statements:

Statement I : A mixture of an alkyl halide and aryl halide gives an alkylarene when treated with sodium in dry ether is called Fittig reaction.

Statement II: For the preparation of 2, 3-Dimethylbutane by Wurtz reaction, the alkyl halide needed as substrate is isopropyl chloride.

In the light of above statements, choose the **correct** option.

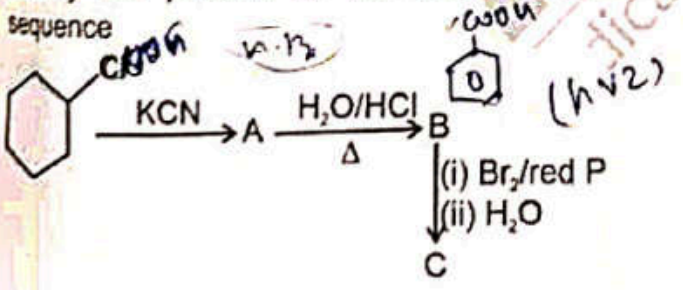
(1) Statement I is incorrect but Statement II is correct

(2) Both statement I and statement II are correct

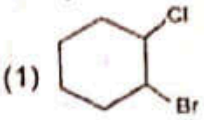
(3) Both statement I and statement II are incorrect

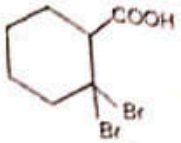
(4) Statement I is correct but Statement II is incorrect

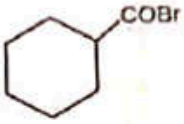
Identify the product of the following reaction sequence

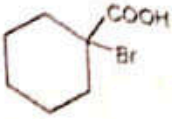


The product 'C' is

(1) 

(2) 

(3) 

(4) 

Product (P) formed in reaction given below is



(1) Propanal (2) Propanone

(3) Ethanol (4) Propene

Consider the following statements.

Statement I : Tertiary alcohol reacts with Lucas reagent faster than secondary alcohol.

Statement II: Lucas reagent is dil HCl + $ZnCl_2$

In the light of above statements, choose the **correct** option.

(1) Statement I is incorrect but Statement II is correct

(2) Both statement I and statement II are correct

(3) Both statement I and statement II are incorrect

(4) Statement I is correct but Statement II is incorrect

Choose the incorrect statement.

(1) The reaction of phenol with carboxylic acid is reversible

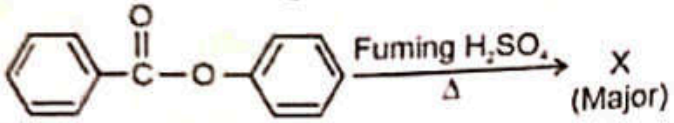
(2) The reaction of phenol with acid chloride is carried out in the presence of pyridine

(3) Acetylation of m-hydroxybenzoic acid produces aspirin

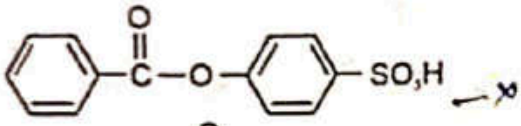
(4) Aspirin possesses analgesic and antipyretic properties

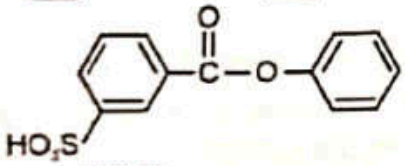
76. IUPAC name of glycerol is
(1) Propane-1,3-diol
(2) 2-Methylpropan-2-ol
(3) Propane-1,2,3-triol
(4) Ethane -1,2-diol

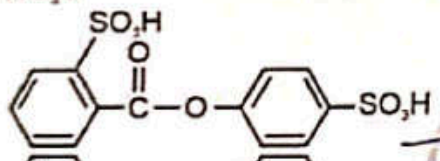
77. Consider the following reactions

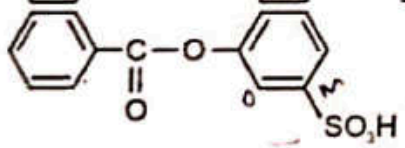


'X' is

(1) 

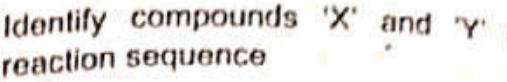
(2) 

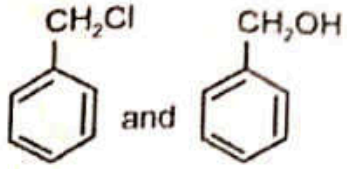
(3) 

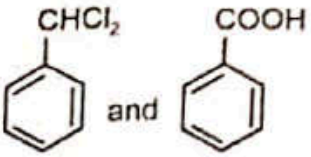
(4) 

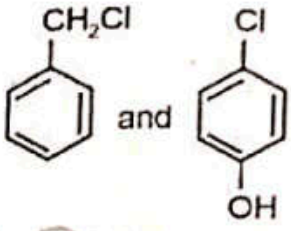
78. Consider the following statements
Statement I: Conformers are rapidly interconvertible and non-separable.
Statement II: Dihedral angle of least stable conformer of ethane is 60°
In the light of above statements, choose the **correct** option.
(1) Statement I is incorrect but Statement II is correct
(2) Both statement I and statement II are correct
(3) Both statement I and statement II are incorrect
(4) Statement I is correct but Statement II is incorrect

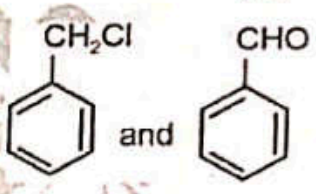
79. Identify compounds 'X' and 'Y' in the give reaction sequence



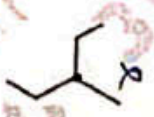
(1) 

(2) 

(3) 

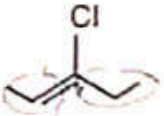
(4) 

80. Which of the following is an optically activ compound?

(1) 

(2) 

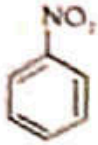
(3) 

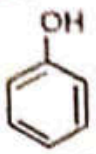
(4) 

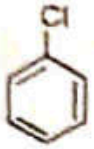
81. Total number of electrons involved in conjugatio in furan  is

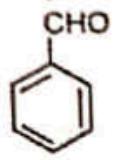
(1) 2
(2) 4
(3) 6
(4) 8

82. The most reactive compound towards electrophilic aromatic substitution reaction is

(1) 

(2) 

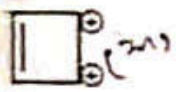
(3) 

(4) 

83. On heating propyne in red hot iron tube at 873 K temperature, product 'P' is formed. The total number of sigma bonds and pi bonds respectively in compound 'P' are

(1) 21 σ bonds and 6 π bonds
(2) 18 σ bonds and 3 π bonds
(3) 18 σ bonds and 6 π bonds
(4) 21 σ bonds and 3 π bonds

84. The aromatic species among the following is

(a) 

(b) 

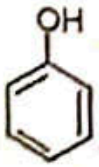
(c) 

(d) 

Choose the correct option.

(1) (a) and (d) only
(2) (a) and (c) only
(3) (a) and (b) only
(4) (b) and (d) only

85. Consider the following reaction

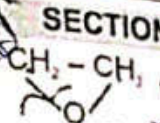


+ A ⟶ B
(Major)

Choose the set showing incorrect match of A and B.

	A	B
(1)	(i) Aq.NaOH + $CHCl_3$ (ii) H^+	Salicylaldehyde
(2)	Zn dust + heat	Benzene
(3)	(i) NaOH (ii) CO_2 (iii) H^+	Salicylic acid
(4)	$K_2Cr_2O_7$, H_2SO_4	Benzoic acid

SECTION-B

86. Reaction of  with RMgX followed by hydrolysis results in the formation of

(1) 

(2) $RCHOHCH_3$

(3) RCH_2CH_2OH

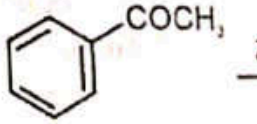
(4) 

87. The number of structural isomeric alcohols molecular formula $C_4H_{10}O$ is

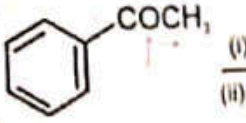
(1) 2 (2) 3
(3) 4 (4) 5

88. Consider the following reactions :

Reaction I:



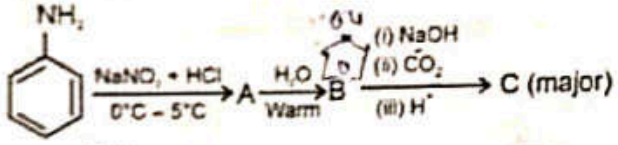
Reaction II:

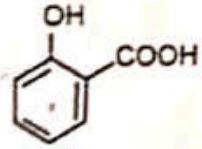


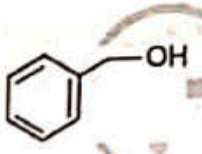
Choose the **correct** option to identify reaction I, reaction II, product P and product Q.

	Reaction I	Reaction II	Product P	Product Q
(1)	Clemmensen reduction	Wolff-Kishner reduction	Toluene	Toluene
(2)	Clemmensen reduction	Wolff-Kishner reduction	Ethyl benzene	Ethyl benzene
(3)	Wolff-Kishner reduction	Clemmensen reduction	Toluene	Ethyl benzene
(4)	Wolff-Kishner reduction	Clemmensen reduction	Ethyl benzene	Toluene

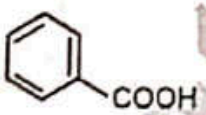
89. Identify major product 'C' in the given reaction sequence



(1) 

(2) 

(3) 

(4) 

90. Read the following assertion (A) and reason (R) and select the **correct** option.

Assertion (A) : 2-chloro-2-methylpropane reacts with sodium ethoxide to form 2-Methylpropene as major product.

Reason (R) : If a tertiary alkyl halide reacts with strong base (sodium ethoxide) then elimination is preferred over substitution.

In the light of above statement, choose the **correct** option.

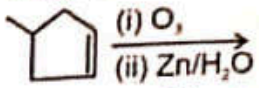
(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

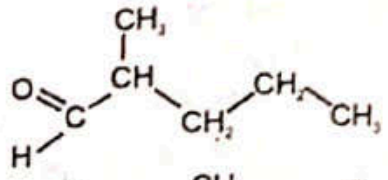
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

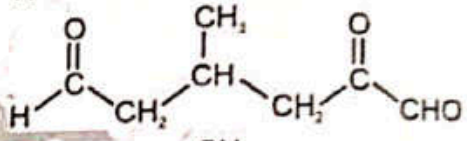
(3) (A) is true but (R) is false

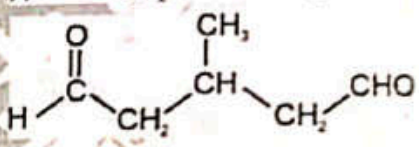
(4) Both (A) and (R) are false

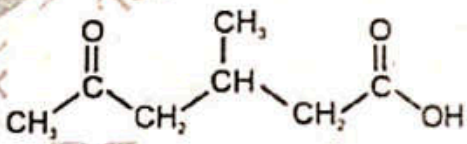
91. The product of the following reaction is



(1) 

(2) 

(3) 

(4) 

92. Consider the following statements regarding methoxy methane

(i) It is a non-polar compound

(ii) It's boiling point is less than that of ethyl alcohol

(iii) It reacts with sodium to evolve hydrogen gas

Choose the **incorrect** statements.

(1) (i) and (ii) only (2) (ii) and (iii) only

(3) (i) and (iii) only (4) (i), (ii) and (iii)

Space for Rough Work

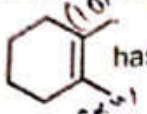
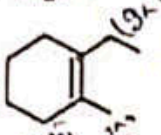
93. Which of the following is the least acidic?

(1) $CH_3 - C \equiv CH$ (2) $CH \equiv CH$

(3) $CH_3 - CH = CH_2$ (4) $CH_3 - CH_2 - CH_3$

94. Consider the following statements:

Statement I : Alkene with higher value of heat of hydrogenation is more stable.

Statement II:  has higher heat of hydrogenation as compared to 

In the light of above statements, choose the **correct** option.

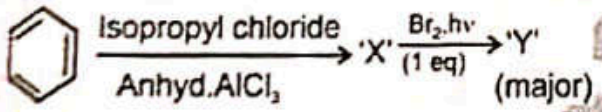
(1) Statement I is incorrect but Statement II is correct

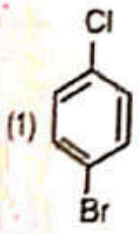
(2) Both statement I and statement II are correct

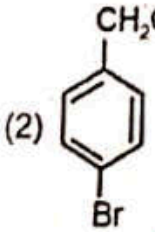
(3) Both statement I and statement II are incorrect

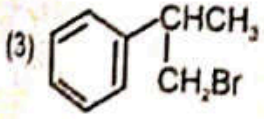
(4) Statement I is correct but Statement II is incorrect

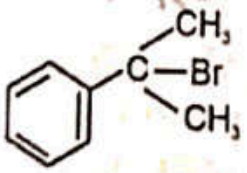
95. The major product 'Y' obtained in the following reaction sequence is



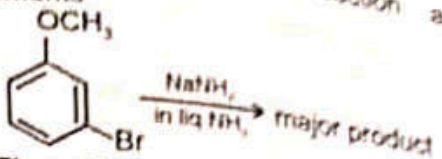
(1) 

(2) 

(3) 

(4) 

96. Consider the following reaction and the statements



(i) The major product formed during the reaction is o-methoxyaniline

(ii) It is an example of nucleophilic substitution

(iii) The intermediate formed during this reaction is benzyne

Choose the **correct** statement.

(1) (i) and (ii) only (2) (iii) only

(3) (ii) and (iii) only (4) (i), (ii) and (iii) only

97. Identify the correct statement with respect to esterification of carboxylic acid.

(a) Esterification of carboxylic acids with alcohols is a kind of nucleophilic acyl substitution

(b) A tetrahedral intermediate is formed

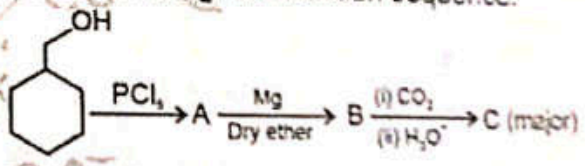
(c) Carboxylic acids are esterified with alcohols or phenols in the presence of concentrated H_2SO_4 as a catalyst

Choose the **correct** option.

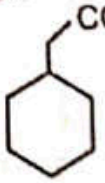
(1) (a) and (b) only (2) (b) and (c) only

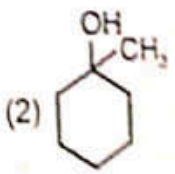
(3) (b) only (4) (a), (b) and (c)

98. Consider the given reaction sequence:

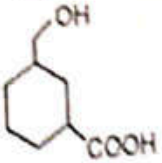


The product C will be

(1) 

(2) 

(3) 

(4) 

99. Given below are the two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : Haloalkanes react with AgCN to form alkyl cyanide as the major product

Reason (R) : AgCN is mainly covalent in nature and nitrogen is free to donate electron pair forming isocyanide as the major product.

In the light of above statement, choose the correct option.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) (A) is false but (R) is true

100. Match the **List-I** of the given process with their respective name reaction given in **List-II**

	List-I		List-II
a.	Conversion of aryl halide into diphenyl	(i)	Finkelstein reaction
b.	Conversion of alkyl chloride into alkyl fluoride	(ii)	Sandmeyer reaction
c.	Conversion of benzene diazonium halide into aryl halide	(iii)	Swarts reaction
d.	Conversion of alkyl bromide into alkyl iodide	(iv)	Fittig reaction

Select the **correct** option.

(1) a(ii), b(iv), c(i), d(iii) (2) a(ii), b(i), c(iv), d(iii)

(3) a(iv), b(i), c(iii), d(ii) (4) a(iv), b(iii), c(ii), d(i)

SECTION-A

101. Select the **mismatched** pair.

(1)	Henking	–	Traced a specific nuclear structure all through spermatogenesis in a few insects
(2)	Punnett square	–	Graphical representation to calculate the probability of all the possible genotypes of offspring in a cross
(3)	Alleles	–	Genes which code for a pair of contrasting traits
(4)	Multiple alleles	–	Exemplified by starch synthesis gene in pea

102. A female affected with haemophilia marries a man whose mother was a carrier for haemophilia. What percentage of male children of this couple will be haemophilic?

(1) 25%

(2) 100%

(3) 75%

(4) 50%

Space for Rough Work

103. Read the following features and choose the **correct** option

(a) Pollen grains are generally surrounded by mucilaginous covering

(b) Female flowers have long pedicel

(c) Male flowers are released on to the surface of water

(d) Found in fresh water

The plant is most likely to be :

(1) Water hyacinth (2) *Vallisneria*

(3) *Zostera* (4) Sea grass

104. The alleles do not show blending and both characters are recovered in F_2 generation of a monohybrid cross. Which law of inheritance is based on above fact

(1) Law of dominance

(2) Law of independent assortment

(3) Law of segregation

(4) Incomplete dominance

105. Cystic fibrosis is a/an

(1) X-linked recessive disorder

(2) Autosomal dominant disorder

(3) Autosomal recessive disorder

(4) X-linked dominant disorder

106. Read the following statements and choose the **correct** option.

Statement A: The ability of a pistil to recognise the pollen followed by its acceptance or rejection is a result of continuous dialogue between pollen and pistil, mediated by chemical components.

Statement B: During post fertilisation embryo development, the zygote gives rise to the proembryo and subsequently to globular, heart shaped and mature embryo.

(1) Only statement A is correct

(2) Only statement B is correct

(3) Both statement A and B are correct

(4) Both statement A and B are incorrect

107. Which of the following traits of pea plant is expressed only in homozygous condition?

(1) Green pod colour

(2) Axial flower position

(3) Violet flower colour

(4) Constricted pod shape

108. Gregor Johann Mendel worked on *Pisum sativum* for ____

(1) 1855 – 1863 (2) 1858 – 1867

(3) 1856 – 1863 (4) 1850 – 1857

109. How many different types of gametes will be produced by a plant having the genotype AaCCddEeFf?

(1) Two (2) Six

(3) Eight (4) Four

110. Progeny with blood group 'O' can be obtained the couple with genotype

(1) $I^A I^B \times I^A I^B$ (2) $I^A I^A \times I^B I^O$

(3) $I^B I^B \times I^A I^O$ (4) $I^O I^A \times I^O I^B$

111. Read the given Assertion (A) are Reason (R) and choose the **correct** option.

Assertion (A) : In honey bees, development of drones occurs by the means of parthenogenesis

Reason (R) : Male honey bees do not have a grandfather but can have sons.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) Both (A) and (R) are false statements

Space for Rough Work

112. Select the **odd** one out w.r.t. ploidy
(1) Antipodals
(2) Integument of megasporangium
(3) Embryo sac
(4) Synergids

113. Perispermic seeds are found in
(1) Beet (2) Pea
(3) Groundnut (4) Wheat

114. State true (T) or false (F) to the given statements
(A) If apomictic hybrid seeds are sown, no segregation of characters is seen in progeny.
(B) In *Michelia* a multicarpellary syncarpous gynoecium is found
(C) In Solanaceae, Poaceae and Liliaceae families, the pollen grains are viable for months
(D) Germ pores are the prominent apertures present in the exine of pollen grain where sporopollenin is absent

	(A)	(B)	(C)	(D)
(1)	F	T	F	T
(2)	F	T	T	T
(3)	T	F	F	T
(4)	F	F	F	T

115. Which of the following statements are **correct** about Down's syndrome?
(A) Develops due to trisomy of chromosome number 21
(B) Affected individual lack any of the sex chromosome
(C) Physical, psychomotor and mental development is retarded
(D) It is a chromosomal disorder that follows Mendelian laws of inheritance.
In the light of above statement select the **correct** option.
(1) (B), (C) and (D) only (2) (A) and (C) only
(3) (A), (B) and (C) only (4) (A) and (D) only

116. In a Mendel's dihybrid cross for seed colour and seed shape, if 1000 seeds are obtained in the F_2 generation, how many seeds of F_2 are expected to be heterozygous for at least one of the traits?
(1) 562 (2) 250
(3) 750 (4) 625

117. Read the following statements and choose the **correct** option.
Statement A: Thalassemia differs from sickle-cell anaemia as the former is a quantitative problem of synthesising too few globin molecules.
Statement B: In Haemophilia, a single protein that is a part of cascade of proteins involved in blood clotting is affected.
(1) Only statement A is correct
(2) Only statement B is correct
(3) Both statements A and B are correct
(4) Both statements A and B are incorrect

118. Select the **odd** one w.r.t. type of fruits
(1) Apple (2) Strawberry
(3) Banana (4) Cashewnut

119. Which of the following is **not** a true reason for Mendel's success?
(1) He applied statistical methods and mathematical logics for result analysis
(2) A small sampling size gave credibility to his results obtained in experiments
(3) He kept accurate records of his experiments by giving all the details of number and type of organisms
(4) He tried to formulate theoretical explanations for the observed results

Space for Rough Work

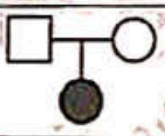
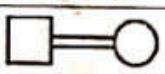
120. Special cellular thickenings present at the micropylar tip in the helper cells are known as
(1) Epiblast
(2) Connective tissue
(3) Egg apparatus
(4) Filiform apparatus

121. A mass of cells having abundant reserve food, enclosed within the protective envelopes of ovule is called
(1) Funicle
(2) Nucellus
(3) Chalaza
(4) Integuments

122. Human skin colour is an example of
(1) Quantitative inheritance
(2) Incomplete dominance
(3) Codominance
(4) Pleiotropy

123. The phenotypic ratio of a monohybrid test cross and of incomplete dominance respectively are
(1) 3 : 1 and 1 : 1 : 1
(2) 1 : 1 and 1 : 2 : 1
(3) 1 : 2 : 1 and 1 : 2 : 1
(4) 1 : 2 : 1 and 3 : 1

124. Match List-I with List-II w.r.t. symbols used in pedigree analysis

	List-I		List-II
a.	Mating between disease affected individuals	(i)	
b.	Consanguineous mating	(ii)	
c.	Sex unspecified	(iii)	
d.	Parents with female child affected with disease	(iv)	 (b)

Choose the **correct** option
(1) a(iv), b(ii), c(iii), d(i)
(2) a(i), b(iii), c(iv), d(ii)
(3) a(iv), b(iii), c(ii), d(i)
(4) a(iii), b(iv), c(i), d(ii)

125. In grasses, epiblast is
(1) Sheild shaped cotyledon, situated towards one side of embryonal axis
(2) Hollow foliar structure enclosing leaf primordia
(3) Remains of second cotyledon
(4) A cylindrical portion below the level of cotyledons which terminates into root tip

126. Which of the following is **not** true about the event unique to angiosperms?
(1) Two types of fusion occurs
(2) One male gamete fuses with egg cell's nucleus
(3) The central cell after triple fusion becomes primary endosperm cell (PEC)
(4) The other male gamete fuses with diploid polar nuclei

127. Select the **correctly** matched pair

(1)	Down's syndrome	–	Small and wrinkled tongue
(2)	*Drosophila melanogaster*	–	Has only 4 morphologically distinct chromosomes
(3)	Snapdragon	–	Flower colour shows codominance
(4)	Polygenic inheritance	–	Bell-shaped normal distribution curve

Space for Rough Work

128. Transfer of pollen grains from the anther to the stigma of another flower of the same plant is known as

(1) Autogamy
(2) Geitonogamy
(3) Xenogamy
(4) Cleistogamy

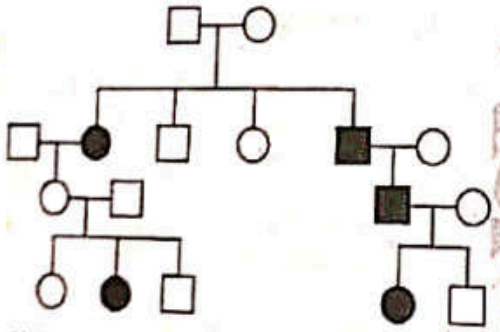
129. In the seed of monocots, an undifferentiated sheath enclosing the root cap is called

(1) Coleorhiza
(2) Coleoptile
(3) Aleurone layer
(4) Scutellum

130. Select the **correct** statement about test cross

(1) It is self-crossing of F_1 hybrid
(2) It involves crossing between two genotypes with dominant traits
(3) It was performed by Mendel to predict linkage between two traits
(4) Involves crossing of an organism with dominant phenotype and the one with recessiv phenotype

131. In the below given pedigree chart, the mut[illegible]rait is shaded black. The gene responsible for the trait is



(1) X-linked recessive trait
(2) X-linked dominant trait
(3) Autosomal recessive trait
(4) Autosomal dominant trait

132. State true (T) or false (F) for the given statements

(A) Morgan obtained 37.2% recombinants in F_2 generation when he crossed yellow bodied white-eyed female with brown bodied red-eyed male.

(B) According to the Mendel's law of inheritance which is based on the results of dihybrid crosses, segregation of one pair of factors independent of the other pair and the gametes must carry all the possible combinations of factors in equal frequency.

	(A)	(B)
(1)	T	T
(2)	T	F
(3)	F	F
(4)	F	T

133. Read the following Assertion (A) or Reason (R) and choose the **correct** option.

Assertion (A) : In *Viola* and *Commelina*, seed setting is assured even in the absence of pollinating agents.

Reason (R) : *Viola* and *Commelina* have cleistogamous flowers that are invariably autogamous

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(3) Both (A) and (R) are false statements
(4) (A) is true but (R) is false

Space for Rough Work

134. Match List-I with List-II

	List-I		List-II
a.	*Zostera*	(i)	Can have colourful flowers and presence of nectaries
b.	Bamboo	(ii)	Insect pollinated plant
c.	*Yucca*	(iii)	Wind pollinated plant
d.	Water lily	(iv)	Marine plant showing hydrophily

Choose the **correct** option.

(1) a(ii), b(i), c(iii), d(iv) (2) a(iv), b(iii), c(i), d(ii)

(3) a(i), b(iv), c(ii), d(iii) (4) a(i), b(ii), c(iii), d(iv)

135. Identify the anther wall layer which surrounds the sporogenous tissue and possesses dense cytoplasm in its cells.

(1) Middle layer (2) Epidermis

(3) Tapetum (4) Endothecium

SECTION-B

136. Select the **incorrect** statement(s) with respect to phenylketonuria.

(A) It is an example of pleiotropy

(B) Phenotypic expression is characterised by mental retardation and increase in hair and skin pigmentation

(C) It is an autosomal dominant trait

(D) Tyrosine is accumulated in the brain due to the lack of an enzyme that converts tyrosine to phenylalanine

(E) This is an inborn error of metabolism

(1) (A), (E) and (C) only

(2) (A) and (D) only

(3) (C), (D) and (E) only

(4) (B), (C) and (D) only

137. Read the following statements and choose the **correct** option.

Statement A: The vegetative cell of pollen is spindle shaped with dense cytoplasm and nucleus

Statement B: Sporopollenin can withstand high temperature but not strong acids and alkali.

(1) Both statements A and B are correct

(2) Both statements A and B are incorrect

(3) Only statement A is correct

(4) Only statement B is correct

138. Select the **correct** statements.

(a) Multiple alleles can be found only when population studies are made.

(b) Dominance is not an autonomous feature of a gene or the product that it has information for

(c) Sickle cell anaemia is caused by substitution of valine by glutamic acid at the sixth position in globin chain of haemoglobin molecule.

(d) α-thalassemia is controlled by a single gene HBB on chromosome 11 of each parent.

(e) The possibility of a female becoming haemophilic is extremely rare.

(1) (a), (c) and (d) only

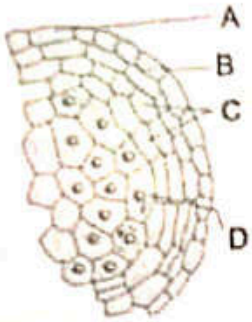
(2) (b), (c) and (d) only

(3) (a), (b) and (c) only

(4) (a), (b) and (e) only

Space for Rough Work

9. Select the **correct** option w.r.t. the figure of anther showing wall layers given below.



(1) 'B' represents the layer having ephemeral cells, which degenerates at maturity

(2) 'A' represents the layer having α-cellulosic fibrous bands in their cells arising from inner tangential wall.

(3) 'D' represents the layer having role in the nourishment of developing pollen grains.

(4) 'C' represents layer which help in dehiscence of anther due to their hygroscopic nature and is always single layered.

140. Select the **incorrect** statement.

(1) The mutant haemoglobin molecule undergoes polymerisation under low O_2 tension, causi the change in the shape of RBC in case of sickle cell anaemia

(2) Law of dominance proposed by Mendel is universally applicable.

(3) A true breeding line shows stable trait inheritance and expression for several generations.

(4) Completely linked genes in a dihybrid cross have F_2 phenotypic ratio of 3 : 1

141. Identify the **incorrect** statement.

(1) The most common type of embryo sac in angiosperms is monosporic.

(2) If the female parent produces unisexual flowers, there is no need for emasculation during artificial hybridisation.

(3) Some reptiles have also been reported as pollinators in some plant species.

(4) In over 60 percent of angiosperms, pollen grains are shed at 3-celled stage.

142. *Drosophila melanogaster* is suitable for experimental genetics because

(a) It could be grown on a simple synthetic medium in the laboratory

(b) The fly has a short life cycle of about two days

(c) A single mating produces hundreds of offsprings

(d) It has many types of hereditary variation that can only be seen under electron microscope

(1) (a), (b) and (c) only (2) (a) and (c) only

(3) (b), (c) and (d) only (4) (a), (b) and (d) only

143. Chromosomal map

(a) Was first formed by Alfred Sturtevant

(b) Is not useful in preparing Genome sequence

(c) Involves use of recombination frequency to deduce the distance between genes on a chromosome

In the light of above statement select the **correct** answer from the given options.

(1) Only (a) and (c) (2) Only (a)

(3) Only (c) (4) Only (b) and (c)

144. Select the **incorrectly** matched pair.

(1)	Micropyle	–	Facilitates entry of O_2 and water during seed germination
(2)	Adventive embryony	–	Can be seen in *Citrus*
(3)	*Orobanche*	–	Parasitic species, containing many tiny seeds
(4)	*Hydrilla*	–	Presence of non-sticky and light weight, dry pollen grains

Space for Rough Work

145. Read the following, Assertion (A) and Reason (R), and choose the **correct** option.

Assertion (A) : Colour blindness occurs in about 8 percent of females and only about 0.4 percent of males

Reason (R) : The gene that lead to red-green colour blindness is present on the autosome.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) Both (A) and (R) are false statements

146. Select the **correct** option about castor

(1) Male and female flowers are produced on two separate plants

(2) Its has non-endospermous seeds

(3) It prevents autogamy but not geitonogamy

(4) Flower are invariably autogamous and ... uce assured seed-set even in absence of pollinators.

147. Match List-I with List-II.

	List-I		List-II
a.	Birds	(i)	Male heterogametic with XO type sex-determination
b.	Honey bee	(ii)	Female heterogametic with ZO type sex-determination
c.	Grasshopper	(iii)	Female heterogametic with ZW type sex-determination
d.	Butterflies	(iv)	Haplo-diploid sex-determination

Choose the **correct** option.

(1) a(i), b(iii), c(ii), d(iv)

(2) a(ii), b(iv), c(iii), d(i)

(3) a(iii), b(iv), c(i), d(ii)

(4) a(ii), b(i), c(iv), d(iii)

148. Pollen grains are

(1) Generally spherical measuring about 25-50 micrometres in diameter

(2) Stored for years in liquid N_2 (– 160°C)

(3) Product of megasporogenesis in sporogenous tissue

(4) Of no taxonomic significance

149. Experimental verification of chromosomal theory of inheritance was done by

(1) T.H. Morgan

(2) Sutton and Boveri

(3) Alfred Sturtevant

(4) Von Tschermak

150. In which of the following genetic disorders, an affected individual has rudimentary ovaries, lacks other secondary sexual characters and has only 45 chromosomes?

(1) Klinefelter's syndrome

(2) Turner's syndrome

(3) Edward syndrome

(4) Down's syndrome

Space for Rough Work

[ZOOLOGY]

SECTION-A

151. All of the following are included under the umbrella of 'Reproductive and Child Health Care programmes' **except**
(1) Bringing up socially conscious healthy families of desired size
(2) Post-natal care of mother and child
(3) Promoting social evils and no sex education in schools
(4) Educating people about the importance of breast feeding

152. A technique in which transfer of an ovum collected from a donor into fallopian tube of another female who cannot produce one, but can provide suitable environment for fertilisation and further development is
(1) ZIFT (2) ICSI
(3) GIFT (4) IUI

153. Select the **incorrect** statement.
(1) According to 2011 census report, the population growth rate of India was 20/10000/year.
(2) IUDs or progestogen-estrogen combinations or progestogens alone are very effective emergency contraceptives.
(3) Inability to conceive or produce children even after 2 years of unprotected sexual co-habitation is called infertility.
(4) Spermicidal foams can be used along with cervical caps to increase contraceptive efficiency.

154. An ART which is employed to form an embryo in the lab in which sperm is directly injected into the ovum is
(1) ICSI (2) IUI
(3) AI (4) GIFT

155. Steroidal contraceptive pills should be taken
(1) Daily for a period of one week during menstruation
(2) Daily for a period of 21 days starting within the first five days of menstrual cycle
(3) Daily for a period of 14 days starting within the first six days of menstrual cycle
(4) Daily for a period of 32 days after menses ends

156. The secretions of which of the following mainly helps in lubrication of penis?
(1) Seminal vesicle
(2) Prostate gland
(3) Bulbourethral gland
(4) Vasa efferentia

157. STIs can spread through
(a) Sharing injection needles with an infected person
(b) Transfusion of blood from an infected person
(c) Infected mother to her foetus
Select the **correct** option.
(1) Only (a) and (b) (2) Only (b) and (c)
(3) (a), (b) and (c) (4) Only (a) and (c)

158. Similar mode of action is shown by
(1) Implants and diaphragm
(2) Multiload 375 and vaults
(3) Oral steroidal contraceptive pills and implants
(4) Cervical caps and Cu7

159. All of the following are correct w.r.t. action of IUDs. **except**
(1) Increase the sperm motility
(2) Make uterus unsuitable for implantation
(3) Suppress the fertilizing capacity of sperms
(4) Make the cervix hostile to sperms

Space for Rough Work

160. In humans, anterior portion of the sperm head is covered by a cap-like structure which is filled with enzymes and is called
(1) Haploid nucleus (2) Acrosome
(3) Middle piece (4) Neck

161. Select the **incorrect** statement.
(1) Infertility can occur due to immunological or psychological disturbances.
(2) Nirodh is a barrier contraceptive used by females to cover cervix.
(3) In amniocentesis, amniotic fluid of the developing foetus is taken out to analyse the foetal cells.
(4) MTPs are considered relatively safe during the first trimester of pregnancy.

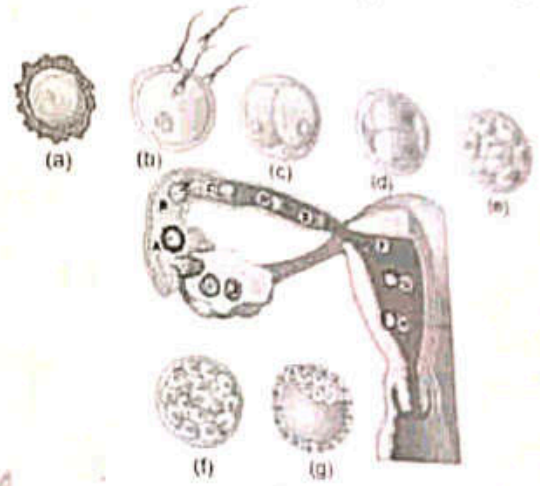
162. Read the following statements.
Statement A: In human females, the part of fallopian tubes closer to ovary is funnel-shaped infundibulum.
Statement B: The last part of oviduct, has a wide lumen and it joins cervix.
Choose the **correct** option.
(1) Both statements A and B are correct
(2) Both statements A and B are incorrect
(3) Only statement A is incorrect
(4) Only statement B is incorrect

163. All of the below given factors can be the reason for population explosion, **except**
(1) Decline in death rate
(2) Decrease in maternal mortality rate
(3) Decrease in infant mortality rate
(4) Decrease in number of people in reproducible age

164. In periodic abstinence, couples avoid from coitus from day ____ of the 30 days menstrual cycle. Select the **correct** option to fill in the blank.
(1) 10 to 17 (2) 12 to 19
(3) 16 to 23 (4) 25 to 28

165. An ideal contraceptive should
(a) Be user – friendly (b) Be easily available
(c) Be irreversible (d) Have least side effects
(e) Interfere with the sexual desire/act of the user
Select the option with only **correct** features.
(1) (a), (b) and (c) (2) (c), (d) and (e)
(3) (b), (d) and (e) (4) (a), (b) and (d)

166. Observe the figure given below.



Choose the **incorrect** option w.r.t. the above given figure.
(1) (e)-Represents the embryo with 8 blastomeres
(2) (g)-Represents blastocyst
(3) (H)-Indicates the event insemination
(4) (e)-Represents morula

167. Select the **incorrect** statement w.r.t. humans.
(1) Vagina is a part of the female accessory ducts.
(2) The spermatids are transformed into spermatozoa by the process called spermiogenesis.
(3) After spermiogenesis, sperm heads become embedded in the interstitial cells.
(4) Vas deferens receives a duct from seminal vesicle and opens into urethra as the ejaculatory duct.

Space for Rough Work

168. Select the **incorrect** match w.r.t. embryonic development of humans.

(1)	End of 3 weeks	–	Foetus develops limbs and digits
(2)	During the 5th month	–	First movement of foetus
(3)	By the end of 4 weeks	–	Embryo's heart is formed
(4)	By the end of 1st trimester	–	Major organ systems are formed

169. The hormone 'X' is secreted by the developing follicles as well as by the temporary endocrine gland formed during the luteal phase. 'X' attains the peak twice in a menstrual cycle. Identify 'X' and choose the **correct** option.

(1) Estrogen (2) Progesterone
(3) LH (4) Relaxin

170. The number of second meiotic divisions required for the formation of 64 spermatids will be

(1) 64 (2) 16
(3) 32 (4) 24

171. Select the **correct** statement.

(1) Ureter acts as the urinogenital duct in human males.
(2) Each seminiferous tubule is lined on its outside by spermatogonia and Sertoli cells.
(3) Cervical canal along with uterus forms the birth canal.
(4) The presence or absence of hymen is not a reliable indicator of sexual experience.

172. In test tube baby programme, after *in vitro* fertilization, the early embryo with less than 8 blastomeres will be transferred to

(1) An inverted pear-shaped structure which is attached to the pelvic wall by ligaments
(2) A tiny finger-like structure which lies at the upper junction of two labia minora
(3) A duct which is about 10-12 cm long and extends from the periphery of each ovary
(4) A fleshy fold of tissue which extends down from the mons pubis

173. **Assertion (A):** Not all copulations lead to fertilisation and pregnancy.

Reason (R): Fertilisation can only occur if the ovum and the sperms are transported simultaneously to the ampullary region of the fallopian tube.

In the light of above statements, select the **correct** option.

(1) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(2) Both (A) and (R) are true and (R) is the correct explanation of (A)
(3) (A) is true; (R) is false
(4) Both (A) and (R) are false

174. Choose the **odd** one w.r.t. events of menstrual cycle related with uterus in human females.

(1) Menstrual phase
(2) Secretory phase
(3) Luteal phase
(4) Proliferative phase

175. During fertilisation, a sperm comes in contact with the 'X' layer of the ovum and induces changes in the membrane that blocks the entry of additional sperms. Select the **correct** option w.r.t. 'X'.

(1) 'X' is a glycoprotein layer present between the plasma membrane of oocytes and corona radiata
(2) 'X' is corona radiata
(3) 'X' is a cellular layer formed by the secretions of Graafian follicle
(4) 'X' is a cellular layer developed in the secondary follicle

176. Which among the following can be administered in fertile human female to inhibit ovulation and implantation as well as alter the quality of cervical mucus?
(1) Luteinizing hormone
(2) Oxytocin
(3) Progesterone-estrogen combination
(4) Relaxin

177. Each primary oocyte in human female gets surrounded by a layer of granulosa cells and is called
(1) Secondary follicle (2) Tertiary follicle
(3) Graafian follicle (4) Primary follicle

178. In humans, the enlarged end of male copulatory organ is covered by a loose fold of skin called
(1) Glans penis (2) Foreskin
(3) Hymen (4) Urethral meatus

179. Match column I and column II w.r.t. layers of uterine wall in human females.

	Column I		Column II
a.	Endometrium	(i)	Glandular and highly vascularised
b.	Perimetrium	(ii)	Thick layer of muscles with uninucleated fibres
c.	Myometrium	(iii)	Outermost layer

Choose the **correct** option.
(1) a-(ii), b-(i), c-(iii) (2) a-(i), b-(ii), c-(iii)
(3) a-(iii), b-(i), c-(ii) (4) a-(i), b-(iii), c-(ii)

180. Select the structure that helps in maintaining the low temperature of testes which is necessary for spermatogenesis in human males
(1) Urethra
(2) Testicular lobules
(3) Scrotum
(4) Ureter

181. Choose the unpaired accessory gland present in the male reproductive system of humans.
(1) Penis (2) Seminal vesicle
(3) Prostate gland (4) Bulbourethral gland

182. In human males, the sperms produced are released from the seminiferous tubules into an extra-testicular duct called _____, which structurally leads to vas deferens.
Select the **correct** option to fill in the blank.
(1) Rete testis
(2) Vasa efferentia
(3) Epididymis
(4) Ureter

183. Cells in human testes which secrete and synthesise androgens are located in
(1) Rete testis
(2) Intra-tubular spaces
(3) Seminiferous tubules
(4) Interstitial spaces

184. In human females, oogenesis is initiated during/at
(1) Puberty
(2) Embryonic development
(3) Birth
(4) Menopause

185. To achieve total reproductive health as a social goal, 'family planning' programmes were initiated in India in the year
(1) 1971 (2) 1961
(3) 1951 (4) 1981

SECTION-B

186. Completion of second meiotic division of secondary oocyte occurs
(1) Before puberty
(2) Before insemination
(3) During fusion of ovum and sperm
(4) During embryonic stage

187. If terminal method of contraception is performed in a human male, then
(1) Spermatogenesis will cease
(2) Ejaculate will contain defective sperms
(3) Level of androgens will get reduced in the blood
(4) Semen will be without sperms

188. Test tube baby programme' that involves *in-vitro* fertilisation includes
(1) AI (2) ZIFT
(3) IUI (4) GIFT

189. All of the following are various complications due to STIs, **except**
(1) Conception (2) Still births
(3) Ectopic pregnancies (4) Infertility

190. How many contraceptives given in the box below are reusable?

IUDs, Nirodh, Cervical caps, Vaults, Diaphragms

Select the **correct** option.
(1) Three (2) Four
(3) Five (4) Two

191. If a male ejaculates about 300 million sperms during coitus, then for normal fertility, at least 60% sperms must have normal shape and size and at least ____ of them must show vigorous motility.
Select the **correct** option to fill in the blank.
(1) 20% (2) 30%
(3) 40% (4) 60%

192. How many statements given below is/are **incorrect** w.r.t. menstrual cycle in human females?
(a) Both LH and FSH attain a peak level in the middle of the 28 days of cycle.
(b) No new follicles develop in the luteal phase as levels of gonadotropins are low.
(c) The level of progesterone peaks twice during luteal phase.
(d) The menstrual phase is followed by the proliferative phase.
Select the **correct** option.
(1) Four (2) Three
(3) Two (4) One

193. All of the given statements are **incorrect** w.r.t. human reproduction, **except**
(1) Differentiation of male gamete occurs before reduction division
(2) The secondary follicles consists of single layer of granulosa cells, new theca and fluid filled cavity called antrum
(3) The primary follicle is characterised by the presence of zona pellucida
(4) Primary oocyte completes its first meiotic division within the tertiary follicle.

194. In functional mammary gland, several mammary ducts join to form a wider mammary ampulla which is connected to ____ through which milk is sucked out.
Select the **correct** option to fill the blank.
(1) Mammary lobe (2) Mammary alveolus
(3) Mammary duct (4) Lactiferous duct

195. From the below given list, how many hormones are secreted by placenta?
(a) hCG (b) hPL
(c) Estrogen (d) Progesterone
(e) Prolactin
Choose the correct option.
(1) Four (2) Five
(3) Two (4) Three

196. Which of the following STIs is caused by virus and cannot be treated with antibiotics?
(1) Genital warts (2) Chlamydiasis
(3) Syphilis (4) Gonorrhoea

Space for Rough Work